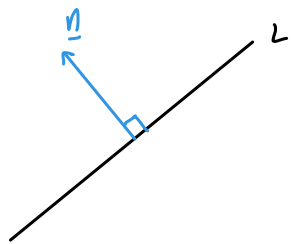
NORMALS



$$\underline{d} = \begin{pmatrix} d_1 \\ d_2 \end{pmatrix}$$

$$\underline{n} = \begin{pmatrix} d_1 \\ -d_2 \end{pmatrix}$$

$$\underline{d} \cdot \underline{n} = d_1 d_2 - d_2 d_1 = 0$$

$\underline{d}$ and $\underline{n}$ are perpendicular

$\underline{n}$ is <u>NOT</u> unique → every vector parallel to $\underline{n}$ is also a normal vector of line L.

$\underline{n}$ is a direction vector for <u>any</u> line which is perpendicular to $\underline{n}$.

REMEMBER: This rule only applies in 2 dimensions. There are an infinite # of directions which are perpendicular to a line in 3D

Problem Find, in slope-intercept form, the equation of a line which passes through the point (8, 4) and has normal vector $9\mathbf{i} + 4\mathbf{j}$

Normal vector : $\underline{n} = \begin{pmatrix} 9 \\ 4 \end{pmatrix}$

Direction vector : $\underline{d} = \begin{pmatrix} 4 \\ -9 \end{pmatrix}$

check : $\underline{n} \cdot \underline{d} = \begin{pmatrix} 9 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -9 \end{pmatrix} = 36 - 36 = 0$

THIS IS CORRECT

Equation of line : $\underline{r} = \begin{pmatrix} 8 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -9 \end{pmatrix}, \quad \lambda \in \mathbb{R}$

TWO LINES

Equations for multiple lines need to be specified → use L_n nomenclature

e.g. $L_1 : \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ 3 \\ 8 \end{pmatrix}$ ← equation of first line

$L_2 : \begin{pmatrix} 3 \\ 5 \\ 8 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 6 \\ 5 \end{pmatrix}$ ← equation of second line

Q: Why do we need to switch from λ to μ?

In 2D, two lines can be:

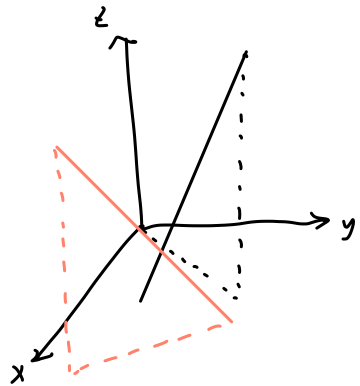
- parallel
- intersecting
- coincident

PARALLEL: direction vectors are scalar multiples of each other

INTERSECTING: there is at least one point which satisfies both equations

COINCIDENT: parallel <u>and</u> intersecting (i.e. describe the same line)

In 3D, there is an additional possibility: <u>SKEW</u>



Skew lines neither intersect nor are parallel.

Each pair of lines must follow one of these 4 relations.

	PARALLEL	NOT PARALLEL
INTERSECTING	COINCIDENT	INTERSECTING
NOT INTERSECTING	PARALLEL	SKEW

Problem Show that the lines given by the equations below are skew:

$$L_1 : \frac{x-1}{3} = \frac{y+2}{2} = \frac{z}{1} \qquad L_2 : \frac{x+3}{-2} = \frac{y}{3} = \frac{z-2}{4}$$

Cartesian form $\rightarrow$ parametric form

L_1

$$\boxed{\begin{aligned} x &= 1 + 3\lambda \\ y &= -2 + 2\lambda \\ z &= \lambda \end{aligned}}$$

L_2

$$\boxed{\begin{aligned} x &= -3 - 2\mu \\ y &= 3\mu \\ z &= 2 + 4\mu \end{aligned}}$$

For skew, we need:

- NOT PARALLEL
- NOT INTERSECTING

PARALLEL?

Direction vectors: $\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix}$ $\quad 3 \cdot c = -2 \Rightarrow c = -\frac{2}{3}, \quad 2 \cdot c = -\frac{4}{3} \neq 3$

Direction vectors are not scalar multiples $\Rightarrow$ NOT parallel

INTERSECTING?

$1 + 3\lambda = -3 - 2\mu \Rightarrow \mu = \dfrac{4 + 3\lambda}{-2} = -2 - \frac{3}{2}\lambda \rightarrow -2 - \frac{3}{2}\lambda = -\frac{2}{3} - \frac{2}{3}\lambda \Rightarrow -12 - 9\lambda = -4 - 4\lambda$

$-2 + 2\lambda = 3\mu \Rightarrow \mu = -\frac{2}{3} + \frac{2}{3}\lambda \rightarrow$

$\Downarrow$

$5\lambda = -8 \Rightarrow \lambda = -\frac{8}{5}$

Sub into (z) $\Rightarrow z = -\frac{8}{5}$ and $z = 2 + 4\left(\frac{-8}{5}\right) = -\frac{22}{5} \neq -\frac{8}{5}$

No points of intersection $\Rightarrow$ NOT intersecting

Not parallel and not intersecting $\Rightarrow$ SKEW

POINT OF INTERSECTION

For POI, equate the right hand sides of the vector equations and solve component-wise.

$$L_1: \underline{r} = \begin{pmatrix}1\\2\\3\end{pmatrix} + \lambda\begin{pmatrix}1\\0\\1\end{pmatrix}$$

$$L_2: \underline{r} = \begin{pmatrix}0\\2\\2\end{pmatrix} + \mu\begin{pmatrix}2\\1\\-1\end{pmatrix}$$

$$\begin{pmatrix}1\\2\\3\end{pmatrix} + \lambda\begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}0\\2\\2\end{pmatrix} + \mu\begin{pmatrix}2\\1\\-1\end{pmatrix}$$

$$1+\lambda = 0+2\mu \Rightarrow \lambda = 2\mu - 1$$

$$2+0\lambda = 2+\mu$$

$$3+\lambda = 2-\mu \Rightarrow \lambda = -\mu - 1$$

$$-\mu - 1 = 2\mu - 1 \Rightarrow \mu = 0$$

$$\lambda = -\mu - 1 = 0 - 1 = -1$$

$$3 + (-1) = 2 - (0)$$

$$2 = 2$$

POINT OF INTERSECTION EXISTS

$$\underline{r} = \begin{pmatrix}1\\2\\3\end{pmatrix} + \lambda\begin{pmatrix}1\\0\\1\end{pmatrix}$$

$$P: \underline{r} = \begin{pmatrix}1\\2\\3\end{pmatrix} + (-1)\begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}0\\2\\2\end{pmatrix}$$

Problem For the line L with equation given below, find:

$$L : \mathbf{r} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k} + \lambda(-\mathbf{i} + \mathbf{j} + 3\mathbf{k}), \lambda \in \mathbb{R}$$

(a) the value of $a \in \mathbb{R}$ for which the point $T(-2,4,a)$ lies on the line
(b) the equation of the line parallel to L and passing through point $P(0,2,5)$

(a) $\underset{\sim}{r} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$

$\left.\begin{array}{l} x = 3 - \lambda = -2 \Rightarrow \lambda = 5 \\ y = -1 + \lambda = 4 \Rightarrow \lambda = 5 \end{array}\right\}$ agreement $\rightarrow$ could use either

$a = 2 + 3\lambda = 2 + 3(5) = 17$

(b) Direction vector: $\underline{d} = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$

Parallel, so we can use the same direction vector

Passes through $P(0,2,5)$

$L : \underset{\sim}{r} = \begin{pmatrix} 0 \\ 2 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$